

AERO40003 Computing and Numerical Methods 1

Part 2: Numerical Methods

Computing Lab 3

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Problem 3.1

Consider a function $f(x) = ax^5 + bx^2 + cx$ where a, b, c take non-zero values of your choice. Compute (analytically) the integral:

$$\int_{-1}^1 f(x) dx = \left[\frac{a}{6}x^6 + \frac{b}{3}x^3 + \frac{c}{2}x^2 \right]_{-1}^1 = \frac{2}{3}b \quad (1)$$

stating clearly what your choice of values for a, b, c is.

Write a piece of code in Matlab to find an approximation to the integral using the Simpson rule. Vary the number of intervals and observe how the error between the Simpson estimate and the analytical answer varies.

For this particular $f(x)$, why does the Simpson rule do so well?

Now repeat the same exercise, this time for the function $g(x) = ax^6 + bx^2 + cx$ where a, b, c again take non-zero values of your choice. Again vary the number of intervals and observe how the error between the Simpson estimate and the analytical answer varies.

Now write another piece of code in Matlab to find an approximation to:

$$\int_{-1}^1 g(x) dx = \left[\frac{a}{7}x^7 + \frac{b}{3}x^3 + \frac{c}{2}x^2 \right]_{-1}^1 = \frac{2}{7}a + \frac{2}{3}b \quad (2)$$

using the Gauss-Legendre quadrature rule. Progressively increase the number of quadrature points and at each stage check the error between your Gauss-Legendre integral and the analytical one. Compare your findings with the performance of the Simpson rule.

Problem 3.2

Consider the integral

$$\int_0^1 e^{-x^2} dx \quad (3)$$

Which has an exact result 0.746824

Evaluate the integral by writing a code in Matlab that uses a Gauss-Legendre quadrature with 3 quadrature points. Note that the limits of the integral required are not the standard domain $-1 \leq x \leq 1$ required by the Gauss-Legendre quadrature scheme

$$\int_{-1}^1 f(t) dt = \sum_{j=1}^n A_j f_j \quad (4)$$

and so you require a change of variable. You may also want to vary the number of quadrature points and observe the accuracy of your answer.

How do your results compare with the analytical value and with values obtained using another method of numerical integration, e.g. Mid-Point Rule, Trapezium and/or Simpson's (1/3 or 3/8) rule?

Problem 3.1

Use Simpson rule

$f(x)=ax^5+bx^2+cx$ at range $(-1, 1)$

```
% Housekeeping
clear
clc

% Define the value of a,b and c
a=input('Define the value of a');
b=input('Define the value of b');
c=input('Define the value of c');

% Define number of the interval n
n=input('Define number of the interval (must be even) ');

% Calculate the interval width
h=(1-(-1))/n;

% Simpson rule
f=(h/3)*(a*(-1)^5+b*(-1)^2+c*(-1));
j=0;

for i=-1+h:h:1-h
    j=j+1;
    if mod(j,2)==0
        f=f+(h/3)*2*(a*i^5+b*i^2+c*i);
    else
        f=f+(h/3)*4*(a*i^5+b*i^2+c*i);
    end
end

f=f+(h/3)*(a*1^5+b*1^2+c*1);

% Output
A=['By calculation, the result is ',num2str(2*b/3)];
B=['By Simpson rule, the result is ',num2str(f)];
disp(A)
```

By calculation, the result is 23.3333

```
disp(B)
```

By Simpson rule, the result is 23.3333

$f(x)=ax^6+bx^2+cx$ at range $(-1, 1)$

```
% Housekeeping
```

```

clear
clc

% Define the value of a,b and c
a=input('Define the value of a');
b=input('Define the value of b');
c=input('Define the value of c');

% Define number of the interval n
n=input('Define number of the interval (must be even) ');

% Calculate the interval width
h=(1-(-1))/n;

% Simpson rule
f=(h/3)*(a*(-1)^6+b*(-1)^2+c*(-1));
j=0;

for i=-1+h:h:1-h
    j=j+1;
    if mod(j,2)==0
        f=f+(h/3)*2*(a*i^6+b*i^2+c*i);
    else
        f=f+(h/3)*4*(a*i^6+b*i^2+c*i);
    end
end

f=f+(h/3)*(a*1^6+b*1^2+c*1);

% Output
A=['By calculation, the result is ',num2str(2*a/7+2*b/3)];
B=['By Simpson rule, the result is ',num2str(f)];
disp(A)

```

By calculation, the result is 3.5238

```
disp(B)
```

By Simpson rule, the result is 3.53

Use Gauss-Legendre quadrature rule

$f(x)=ax^6+bx^2+cx$ at range $(-1, 1)$

```

% Housekeeping
clear
clc

% Define the value of a,b and c
a=input('Define the value of a');

```

```

b=input('Define the value of b');
c=input('Define the value of c');

% Define the degree N
N=input('Define the degree N');

% Define the value x
syms x

% Legendre polynomials
eq=(x^2-1)^N;

for i=1:N
    eq=diff(eq);
end

eq=1/(2^N*factorial(N))*eq;

% Find ni
ni=solve(eq,x);

% Gauss-Legendre quadrature rule
for j=1:N
    eq2=diff(eq);
    x=ni(j);
    eq3=subs(eq2);
    wi(j)=2/(1-ni(j)^2)*(eq3)^-2;
end

f=0;

for j=1:N
    f=f+wi(j)*(a*ni(j)^6+b*ni(j)^2+c*ni(j));
end

% Output
A=['By calculation, the result is ',num2str(2*a/7+2*b/3)];
disp(A)

```

By calculation, the result is 3.5238

```

B=['By Gauss-Legendre quadrature rule, the result is ',num2str(double(f))];
disp(B)

```

By Gauss-Legendre quadrature rule, the result is 3.3867

Problem 3.2

Use Gauss-Legendre quadrature rule

$f(x)=e^{-x^2}$ at range (0, 1)

```

% Housekeeping
clear
clc

% Define the degree N
N=input('Define the degree N');

% Define the value x and t
syms x
syms t
lt=(0*1-1*(-1))/(1-(-1))+(1-0)/(1-(-1))*t;

% Legendre polynomials
eq=(x^2-1)^N;

for i=1:N
    eq=diff(eq);
end

eq=1/(2^N*factorial(N))*eq;

% Find ni
ni=solve(eq,x);

% Gauss-Legendre quadrature rule
syms a

for j=1:N
    eq2=diff(eq);
    x=lt;
    eq3=subs(eq2);
    t=solve(lt==ni(j));
    eq4=subs(eq3);
    wi(j)=2/(1-ni(j)^2)*(eq4)^-2;
end

f=0;

for j=1:N
    f=f+wi(j)*exp(-ni(j)^2);
end

f=(1-0)/(1-(-1))*f;

% Output
disp('By calculation, the result is 0.746824')

```

By calculation, the result is 0.746824

```
B=['By Gauss-Legendre quadrature rule, the result is ',num2str(double(f))];
```

```
disp(B)
```

By Gauss-Legendre quadrature rule, the result is 0.74934